



UNIVERSIDADE FEDERAL DO RIO GRANDE DO NORTE
CENTRO DE ENSINO SUPERIOR DO SERIDÓ
CURSO DE SISTEMAS DE INFORMAÇÃO



Trabalho de Álgebra Linear

Avaliação 01

VALOR: 10 pontos

Orientador: Prof. Dr. Francisco Márcio Barboza

Questões de Álgebra Linear apresentada como parte da avaliação 01 da disciplina DCT1304 a ser entregue até as 23:59 do dia da primeira Avaliação.

Caicó, AC, junho de 2020

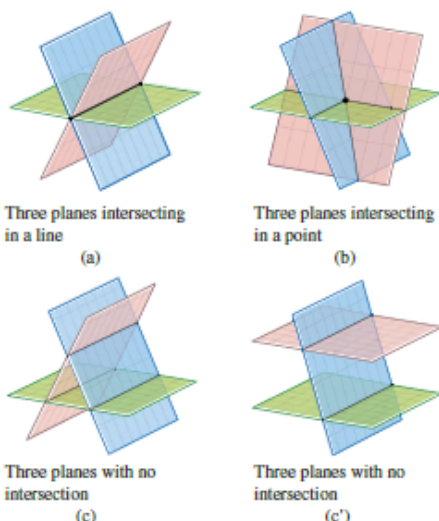
Questões da Prova

Questões - Página 01 (2.5 pontos)

1. Mark each statement True or False. Justify each answer. (If true, cite appropriate facts or theorems. If false, explain why or give a counterexample that shows why the statement is not true in every case.
 - a. Every matrix is row equivalent to a unique matrix in echelon form.
 - b. Any system of n linear equations in n variables has at most n solutions.
 - c. If a system of linear equations has two different solutions, it must have infinitely many solutions.
 - d. If a system of linear equations has no free variables, then it has a unique solution.
 - e. If an augmented matrix $[A \ b]$ is transformed into $[C \ d]$ by elementary row operations, then the equations $A\mathbf{x} = \mathbf{b}$ and $C\mathbf{x} = \mathbf{d}$ have exactly the same solution sets.
 - f. If a system $A\mathbf{x} = \mathbf{b}$ has more than one solution, then so does the system $A\mathbf{x} = \mathbf{0}$.
 - g. If A is an $m \times n$ matrix and the equation $A\mathbf{x} = \mathbf{b}$ is consistent for some $\mathbf{b}$, then the columns of A span $\mathbb{R}^m$.
 - h. If an augmented matrix $[A \ b]$ can be transformed by elementary row operations into reduced echelon form, then the equation $A\mathbf{x} = \mathbf{b}$ is consistent.
 - i. If matrices A and B are row equivalent, they have the same reduced echelon form.
 - j. The equation $A\mathbf{x} = \mathbf{0}$ has the trivial solution if and only if there are no free variables.
 - k. If A is an $m \times n$ matrix and the equation $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$ in $\mathbb{R}^m$, then A has m pivot columns.
 - l. If an $m \times n$ matrix A has a pivot position in every row, then the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b}$ in $\mathbb{R}^m$.
 - m. If an $n \times n$ matrix A has n pivot positions, then the reduced echelon form of A is the $n \times n$ identity matrix.
 - n. If 3×3 matrices A and B each have three pivot positions, then A can be transformed into B by elementary row operations.
 - o. If A is an $m \times n$ matrix, if the equation $A\mathbf{x} = \mathbf{b}$ has at least two different solutions, and if the equation $A\mathbf{x} = \mathbf{c}$ is consistent, then the equation $A\mathbf{x} = \mathbf{c}$ has many solutions.
 - p. If A and B are row equivalent $m \times n$ matrices and if the columns of A span $\mathbb{R}^m$, then so do the columns of B .
 - q. If none of the vectors in the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{R}^3$ is a multiple of one of the other vectors, then S is linearly independent.
 - r. If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly independent, then $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are not in $\mathbb{R}^2$.
 - s. In some cases, it is possible for four vectors to span $\mathbb{R}^5$.
 - t. If $\mathbf{u}$ and $\mathbf{v}$ are in $\mathbb{R}^m$, then $-\mathbf{u}$ is in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$.
 - u. If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^2$, then $\mathbf{w}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$.
 - v. If $\mathbf{w}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, then $\mathbf{u}$ is a linear combination of $\mathbf{v}$ and $\mathbf{w}$.
 - w. Suppose that $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$ are in $\mathbb{R}^5$, $\mathbf{v}_2$ is not a multiple of $\mathbf{v}_1$, and $\mathbf{v}_3$ is not a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. Then $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.
 - x. A linear transformation is a function.
 - y. If A is a 6×5 matrix, the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ cannot map $\mathbb{R}^5$ onto $\mathbb{R}^6$.
 - z. If A is an $m \times n$ matrix with m pivot columns, then the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is a one-to-one mapping.
2. Let a and b represent real numbers. Describe the possible solution sets of the (linear) equation $ax = b$. [Hint: The number of solutions depends upon a and b .]
3. The solutions (x, y, z) of a single linear equation $ax + by + cz = d$ form a plane in $\mathbb{R}^3$ when a, b , and c are not all zero. Construct sets of three linear equations whose graphs (a) intersect in a single line, (b) intersect in a single point, and (c) have no

Questões - Página 02 (2.5 pontos)

points in common. Typical graphs are illustrated in the figure.



4. Suppose the coefficient matrix of a linear system of three equations in three variables has a pivot position in each column. Explain why the system has a unique solution.

5. Determine h and k such that the solution set of the system (i) is empty, (ii) contains a unique solution, and (iii) contains infinitely many solutions.

$$\begin{array}{ll} \text{a. } x_1 + 3x_2 = k & \text{b. } -2x_1 + hx_2 = 1 \\ 4x_1 + hx_2 = 8 & 6x_1 + kx_2 = -2 \end{array}$$

6. Consider the problem of determining whether the following system of equations is consistent:

$$4x_1 - 2x_2 + 7x_3 = -5$$

$$8x_1 - 3x_2 + 10x_3 = -3$$

- Define appropriate vectors, and restate the problem in terms of linear combinations. Then solve that problem.
- Define an appropriate matrix, and restate the problem using the phrase "columns of A ."
- Define an appropriate linear transformation T using the matrix in (b), and restate the problem in terms of T .

7. Consider the problem of determining whether the following system of equations is consistent for all b_1, b_2, b_3 :

$$2x_1 - 4x_2 - 2x_3 = b_1$$

$$-5x_1 + x_2 + x_3 = b_2$$

$$7x_1 - 5x_2 - 3x_3 = b_3$$

- Define appropriate vectors, and restate the problem in terms of $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. Then solve that problem.
- Define an appropriate matrix, and restate the problem using the phrase "columns of A ."

- Define an appropriate linear transformation T using the matrix in (b), and restate the problem in terms of T .

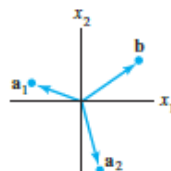
8. Describe the possible echelon forms of the matrix A . Use the notation of Example 1 in Section 1.2.

- A is a 2×3 matrix whose columns span $\mathbb{R}^2$.

- A is a 3×3 matrix whose columns span $\mathbb{R}^3$.

9. Write the vector $\begin{bmatrix} 5 \\ 6 \end{bmatrix}$ as the sum of two vectors, one on the line $\{(x, y) : y = 2x\}$ and one on the line $\{(x, y) : y = x/2\}$.

10. Let $\mathbf{a}_1, \mathbf{a}_2$, and $\mathbf{b}$ be the vectors in $\mathbb{R}^2$ shown in the figure, and let $A = [\mathbf{a}_1 \ \mathbf{a}_2]$. Does the equation $A\mathbf{x} = \mathbf{b}$ have a solution? If so, is the solution unique? Explain.



11. Construct a 2×3 matrix A , not in echelon form, such that the solution of $A\mathbf{x} = \mathbf{0}$ is a line in $\mathbb{R}^3$.

12. Construct a 2×3 matrix A , not in echelon form, such that the solution of $A\mathbf{x} = \mathbf{0}$ is a plane in $\mathbb{R}^3$.

13. Write the *reduced* echelon form of a 3×3 matrix A such that the first two columns of A are pivot columns and $A \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

14. Determine the value(s) of a such that $\left\{ \begin{bmatrix} 1 \\ a \end{bmatrix}, \begin{bmatrix} a \\ a+2 \end{bmatrix} \right\}$ is linearly independent.

15. In (a) and (b), suppose the vectors are linearly independent. What can you say about the numbers $a, \dots, f$? Justify your answers. [Hint: Use a theorem for (b).]

$$\text{a. } \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} b \\ c \\ 0 \end{bmatrix}, \begin{bmatrix} d \\ e \\ f \end{bmatrix} \quad \text{b. } \begin{bmatrix} a \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} b \\ c \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} d \\ e \\ f \\ 1 \end{bmatrix}$$

16. Use Theorem 7 in Section 1.7 to explain why the columns of the matrix A are linearly independent.

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 5 & 0 & 0 \\ 3 & 6 & 8 & 0 \\ 4 & 7 & 9 & 10 \end{bmatrix}$$

17. Explain why a set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ in $\mathbb{R}^5$ must be linearly independent when $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent and $\mathbf{v}_4$ is not in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

18. Suppose $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a linearly independent set in $\mathbb{R}^n$. Show that $\{\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2\}$ is also linearly independent.

Questões - Página 03 (2.5 pontos)

19. Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are distinct points on one line in $\mathbb{R}^3$. The line need not pass through the origin. Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent.

20. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and suppose $T(\mathbf{u}) = \mathbf{v}$. Show that $T(-\mathbf{u}) = -\mathbf{v}$.

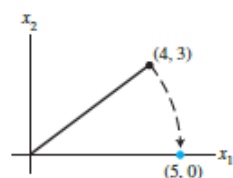
21. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation that reflects each vector through the plane $x_2 = 0$. That is, $T(x_1, x_2, x_3) = (x_1, -x_2, x_3)$. Find the standard matrix of T .

22. Let A be a 3×3 matrix with the property that the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^3$ onto $\mathbb{R}^3$. Explain why the transformation must be one-to-one.

23. A *Givens rotation* is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$ used in computer programs to create a zero entry in a vector (usually a column of a matrix). The standard matrix of a Givens rotation in $\mathbb{R}^2$ has the form

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}, \quad a^2 + b^2 = 1$$

Find a and b such that $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$ is rotated into $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$.



A Givens rotation in $\mathbb{R}^2$.

24. The following equation describes a Givens rotation in $\mathbb{R}^3$. Find a and b .

$$\begin{bmatrix} a & 0 & -b \\ 0 & 1 & 0 \\ b & 0 & a \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2\sqrt{5} \\ 3 \\ 0 \end{bmatrix}, \quad a^2 + b^2 = 1$$

25. A large apartment building is to be built using modular construction techniques. The arrangement of apartments on any particular floor is to be chosen from one of three basic floor plans. Plan A has 18 apartments on one floor, including 3 three-bedroom units, 7 two-bedroom units, and 8 one-bedroom units. Each floor of plan B includes 4 three-bedroom units, 4 two-bedroom units, and 8 one-bedroom units. Each floor of plan C includes 5 three-bedroom units, 3 two-bedroom units, and 9 one-bedroom units. Suppose the building contains a total of x_1 floors of plan A, x_2 floors of plan B, and x_3 floors of plan C.

a. What interpretation can be given to the vector $x_1 \begin{bmatrix} 3 \\ 7 \\ 8 \end{bmatrix}$?

b. Write a formal linear combination of vectors that expresses the total numbers of three-, two-, and one-bedroom apartments contained in the building.

c. [M] Is it possible to design the building with exactly 66 three-bedroom units, 74 two-bedroom units, and 136 one-bedroom units? If so, is there more than one way to do it? Explain your answer.

Questões pares - Página 04 (2.5 pontos)

- Assume that the matrices mentioned in the statements below have appropriate sizes. Mark each statement True or False. Justify each answer.
 - If A and B are $m \times n$, then both AB^T and A^TB are defined.
 - If $AB = C$ and C has 2 columns, then A has 2 columns.
 - Left-multiplying a matrix B by a diagonal matrix A , with nonzero entries on the diagonal, scales the rows of B .
 - If $BC = BD$, then $C = D$.
 - If $AC = 0$, then either $A = 0$ or $C = 0$.
 - If A and B are $n \times n$, then $(A + B)(A - B) = A^2 - B^2$.
 - An elementary $n \times n$ matrix has either n or $n + 1$ nonzero entries.
 - The transpose of an elementary matrix is an elementary matrix.
 - An elementary matrix must be square.
 - Every square matrix is a product of elementary matrices.
 - If A is a 3×3 matrix with three pivot positions, there exist elementary matrices $E_1, \dots, E_p$ such that $E_p \cdots E_1 A = I$.
 - If $AB = I$, then A is invertible.
 - If A and B are square and invertible, then AB is invertible, and $(AB)^{-1} = A^{-1}B^{-1}$.
 - If $AB = BA$ and if A is invertible, then $A^{-1}B = BA^{-1}$.
 - If A is invertible and if $r \neq 0$, then $(rA)^{-1} = rA^{-1}$.
 - If A is a 3×3 matrix and the equation $Ax = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has a unique solution, then A is invertible.
- Find the matrix C whose inverse is $C^{-1} = \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$.
- Let $A = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Show that $A^3 = 0$. Use matrix algebra to compute the product $(I - A)(I + A + A^2)$.
- Suppose $A^n = 0$ for some $n > 1$. Find an inverse for $I - A$.
- Suppose an $n \times n$ matrix A satisfies the equation $A^2 - 2A + I = 0$. Show that $A^3 = 3A - 2I$ and $A^4 = 4A - 3I$.
- Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. These are *Pauli spin matrices* used in the study of electron spin in quantum mechanics. Show that $A^2 = I$, $B^2 = I$, and $AB = -BA$. Matrices such that $AB = -BA$ are said to *anticommute*.
- Let $A = \begin{bmatrix} 1 & 3 & 8 \\ 2 & 4 & 11 \\ 1 & 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 5 \\ 1 & 5 \\ 3 & 4 \end{bmatrix}$. Compute $A^{-1}B$ without computing A^{-1} . [Hint: $A^{-1}B$ is the solution of the equation $AX = B$.]
- Find a matrix A such that the transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 7 \end{bmatrix}$ into $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$, respectively. [Hint: Write a matrix equation involving A , and solve for A .]
- Suppose $AB = \begin{bmatrix} 5 & 4 \\ -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix}$. Find A .
- Suppose A is invertible. Explain why A^TA is also invertible. Then show that $A^{-1} = (A^TA)^{-1}A^T$.
- Let $x_1, \dots, x_n$ be fixed numbers. The matrix below, called a *Vandermonde matrix*, occurs in applications such as signal processing, error-correcting codes, and polynomial interpolation.

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{bmatrix}$$

Given $\mathbf{y} = (y_1, \dots, y_n)$ in $\mathbb{R}^n$, suppose $\mathbf{c} = (c_0, \dots, c_{n-1})$ in $\mathbb{R}^n$ satisfies $V\mathbf{c} = \mathbf{y}$, and define the polynomial $p(t) = c_0 + c_1t + c_2t^2 + \cdots + c_{n-1}t^{n-1}$.

 - Show that $p(x_1) = y_1, \dots, p(x_n) = y_n$. We call $p(t)$ an *interpolating polynomial for the points* $(x_1, y_1), \dots, (x_n, y_n)$ because the graph of $p(t)$ passes through the points.
 - Suppose $x_1, \dots, x_n$ are distinct numbers. Show that the columns of V are linearly independent. [Hint: How many zeros can a polynomial of degree $n - 1$ have?]
 - Prove: "If $x_1, \dots, x_n$ are distinct numbers, and $y_1, \dots, y_n$ are arbitrary numbers, then there is an interpolating polynomial of degree $\leq n - 1$ for $(x_1, y_1), \dots, (x_n, y_n)$."
- Let $A = LU$, where L is an invertible lower triangular matrix and U is upper triangular. Explain why the first column